

M4.1 Unit description

Relative motion; elastic collisions in two dimensions; further motion of particles in one dimension; stability.

M4.2 Assessment information

Prerequisites

A knowledge of the specifications for M1, M2 and M3 and their prerequisites and associated formulae, together with a knowledge of the calculus covered in FP2 and $\int \frac{1}{a^2 + x^2} dx$, is assumed and may be tested.

Examination

The examination will consist of one 1½ hour paper. It will contain about seven questions with varying mark allocations per question which will be stated on the paper. All questions may be attempted.

Calculators

Students are expected to have available a calculator with at least the following keys: +, −, ×, ÷, π, x^2 , $\sqrt{x}$, $\frac{1}{x}$, x^y , $\ln x$, e^x , sine, cosine and tangent and their inverses in degrees and decimals of a degree, and in radians; memory. Calculators with a facility for symbolic algebra, differentiation and/or integration are not permitted.

Formulae

Students are expected to know any other formulae which might be required by the specification and which are not included in the booklet, *Mathematical Formulae including Statistical Formulae and Tables*, which will be provided for use with the paper. Questions will be set in SI units and other units in common usage.

Students will be expected to know and be able to recall and use the following formulae:

$${}_A\mathbf{v}_B = \mathbf{v}_A - \mathbf{v}_B$$

1 Relative motion

What students need to learn:

Relative motion of two particles, including relative displacement and relative velocity.

Problems may be set in vector form and may involve problems of interception or closest approach including the determination of course required for closest approach.

2 Elastic collisions in two dimensions

What students need to learn:

Oblique impact of smooth elastic spheres and a smooth sphere with a fixed surface.

3 Further motion of particles in one dimension

What students need to learn:

Resisted motion of a particle moving in a straight line.

The resisting forces may include the forms $a + bv$ and $a + bv^2$ where a and b are constants and v is the speed.

Damped and/or forced harmonic motion.

The damping to be proportional to the speed. Solution of the relevant differential equations will be expected.

4 Stability

What students need to learn:

Finding equilibrium positions of a system from consideration of its potential energy.

Positions of stable and unstable equilibrium of a system.